

Project Development Phase

No. of Functional Features Included in the Solution

Date	11 July 2026
Team ID	SWTID-2026-9508
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

The table below lists every functional feature implemented in the working solution, mapped to the file(s) where it is implemented.

#	Functional Feature	Description	Implemented In
1	Historical dataset loading & cleaning	Missing value imputation, duplicate removal, dtype correction	Notebooks/EDA_and_Model_Training.ipynb
2	Exploratory Data Analysis	Class balance, distributions, correlation heatmap, outlier detection (8 charts generated)	Notebooks/EDA_and_Model_Training.ipynb, Assets/
3	Feature encoding & scaling	Label encoding of categorical fields, StandardScaler on numeric fields	Notebooks/EDA_and_Model_Training.ipynb
4	Multi-model training	Logistic Regression, Decision Tree, Random Forest, XGBoost all trained and compared	Notebooks/EDA_and_Model_Training.ipynb
5	Model evaluation	Accuracy, Precision, Recall, F1-Score, Confusion Matrix, Classification Report, ROC Curve	Notebooks/EDA_and_Model_Training.ipynb
6	Best-model selection	Automatic comparison table; XGBoost selected by highest F1-Score	Notebooks/EDA_and_Model_Training.ipynb
7	Model persistence	Trained model, encoders, scaler, and feature order saved via joblib	Model/*.pkl
8	Web-based application form	Responsive HTML/Bootstrap-style form for entering applicant details	Flask/templates/index.html
9	Live prediction endpoint	POST /predict route runs the saved model and returns Approved/Rejected	Flask/app.py
10	Confidence score display	Shows the model's predicted probability of approval alongside the result	Flask/app.py, Flask/templates/result.html

#	Functional Feature	Description	Implemented In
11	Input validation	Required-field checks and invalid-category handling with user-facing error messages	Flask/app.py, Flask/templates/index.html
12	Custom error handling	Dedicated 404 page instead of Flask's default error screen	Flask/templates/404.html
13	Logging	Model load status, predictions, and errors logged via Python's logging module	Flask/app.py
14	Consistent visual branding	Shared navigation/footer via Jinja2 template inheritance, banking-themed design	Flask/templates/base.html, Flask/static/style.css

Total functional features implemented: 14, spanning data preprocessing, model training and evaluation, and the full end-to-end web application.